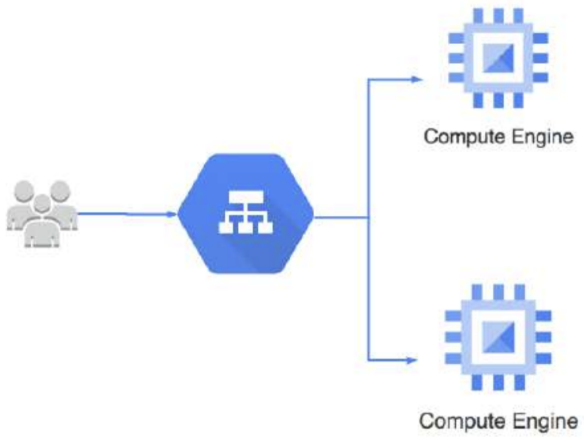
Google Cloud Load Balancing

From DevOps perspective

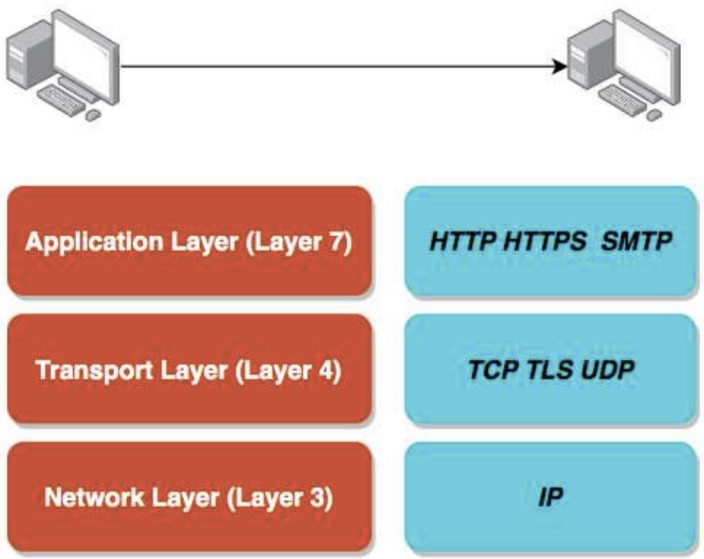
Cloud Load Balancing

- Distributes user traffic across instances of an application in single region or multiple regions
  - **Fully distributed, software defined** managed service
  - Important Features:
    - Health check - Route to healthy instances
      - Recover from failures
    - Auto Scaling
    - Global load balancing with single anycast IP
      - Also supports internal load balancing
- Enables:
  - High Availability
  - Auto Scaling
  - Resiliency



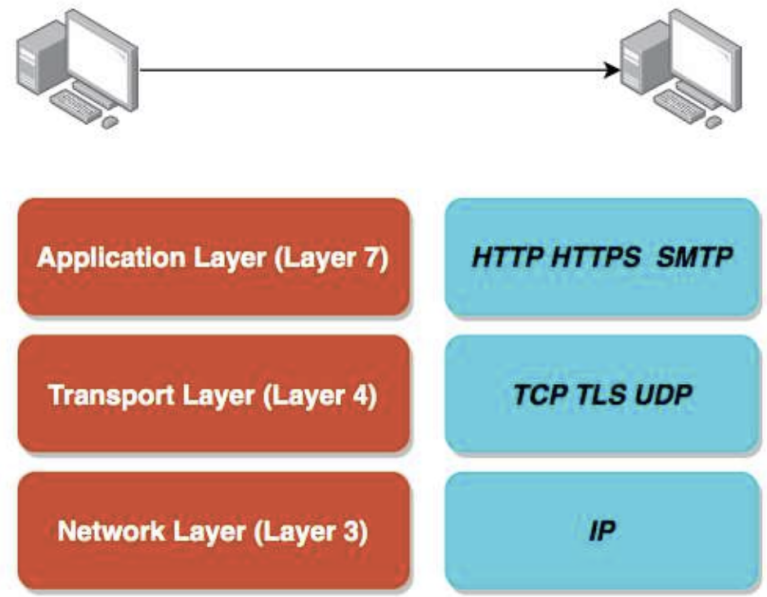
HTTP vs HTTPS vs TCP vs TLS vs UDP

- Computers use protocols to communicate
- Multiple layers and multiple protocols
- **Network Layer** - Transfer bits and bytes
- **Transport Layer** - Are the bits and bytes transferred properly?
- **Application Layer** - Make REST API calls and Send Emails
- (Remember) Each layer makes use of the layers beneath it
- (Remember) Most applications talk at application layer. BUT some applications talk at transport layer directly(high performance).



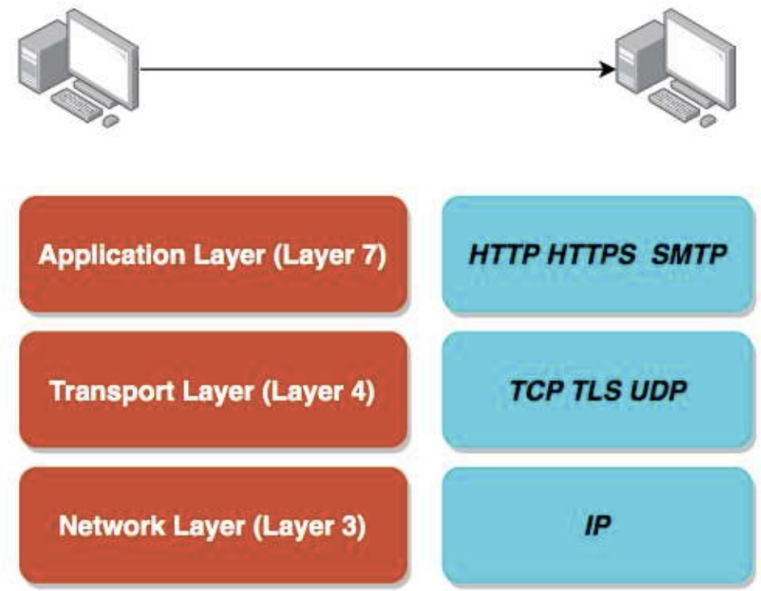
HTTP vs HTTPS vs TCP vs TLS vs UDP

- Network Layer:
  - IP (Internet Protocol): Transfer bytes. Unreliable.
- Transport Layer:
  - TCP (Transmission Control): Reliability > Performance
  - TLS (Transport Layer Security): Secure TCP
  - UDP (User Datagram Protocol): Performance > Reliability
- Application Layer:
  - HTTP(Hypertext Transfer Protocol): Stateless Request Response Cycle
  - HTTPS: Secure HTTP
  - SMTP: Email Transfer Protocol
  - and a lot of others...



HTTP vs HTTPS vs TCP vs TLS vs UDP

- **Most applications** typically communicate at application layer
  - Web apps/REST API(HTTP/HTTPS), Email Servers(SMTP), File Transfers(FTP)
  - All these applications use TCP/TLS at network layer(for reliability)
- **HOWEVER** applications needing high performance **directly** communicate at transport layer:
  - Gaming applications and live video streaming use UDP (sacrifice reliability for performance)
- **Objective**: Understand Big Picture. Its OK if you do not understand all details.



Load balancer types

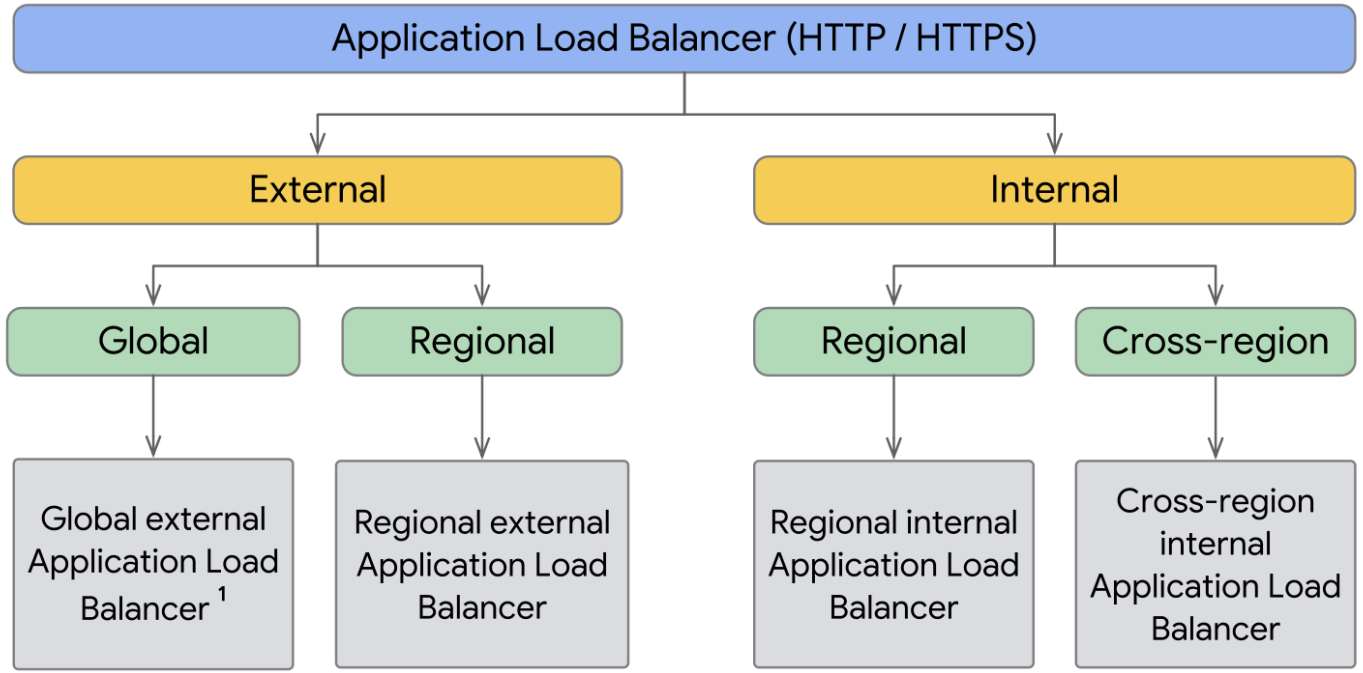
- External Application Load Balancer
- Internal Application Load Balancer
- External passthrough Network Load Balancer
- Internal passthrough Network Load Balancer
- External proxy Network Load Balancer
- Internal proxy Network Load Balancer

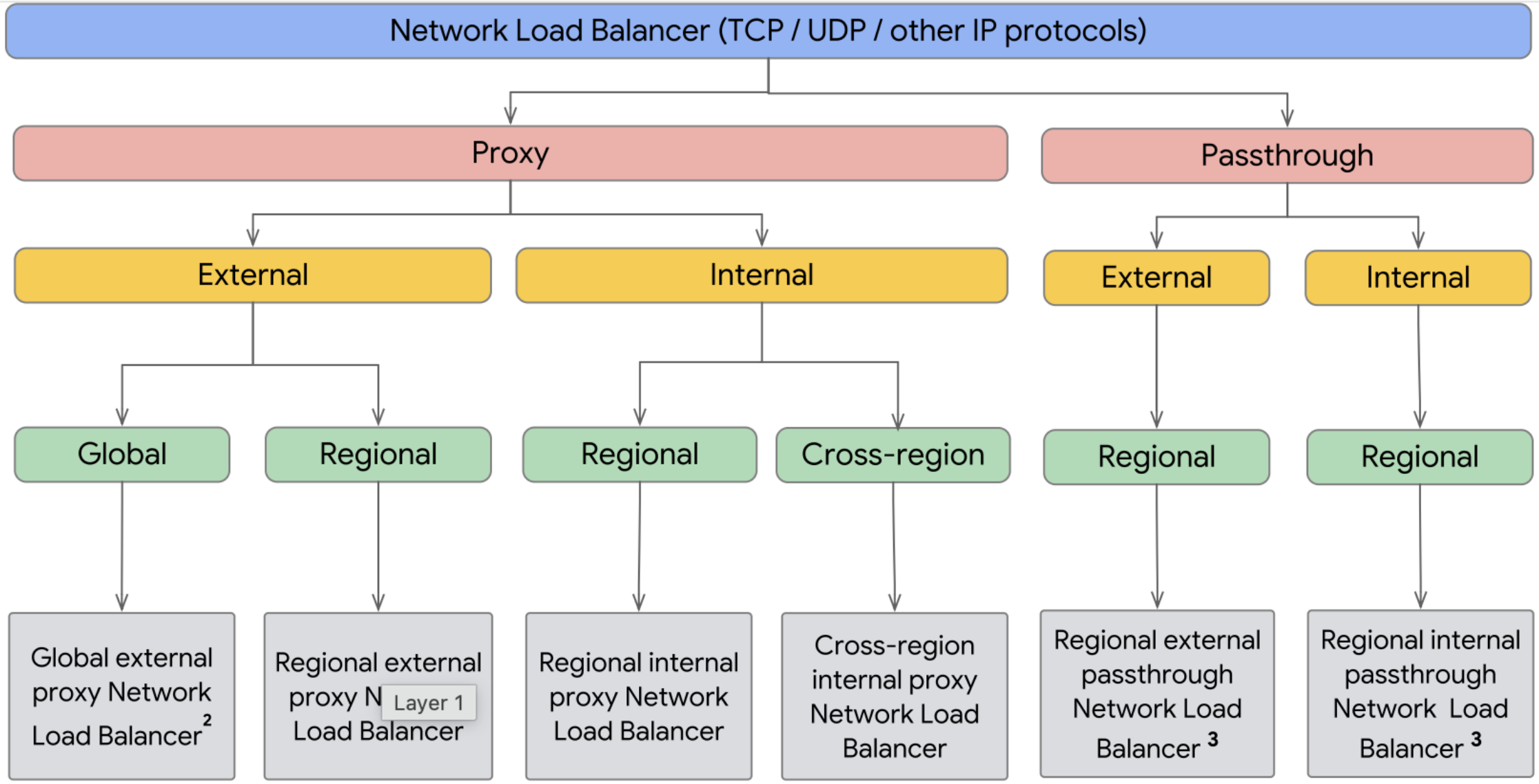
Choose a load balancer

To determine which Cloud Load Balancing product to use, you must first determine the traffic type that your load balancers must handle.

- Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP(S) traffic.
- Choose a proxy Network Load Balancer to implement TCP proxy load balancing to backends in one or more regions.
- Choose a passthrough Network Load Balancer to preserve client source IP addresses, avoid the overhead of proxies, and to support additional protocols like UDP, ESP, and ICMP.

Available Deployment Modes for Cloud Load Balancing




Network Load Balancer (TCP / UDP / other IP protocols)
Proxy
Passthrough
External
Internal
External
Internal
Global
Regional
Regional
Cross-region
Regional
Regional
Global external proxy Network Load Balancer 2
Regional external proxy N Layer 1 Load Balancer
Regional internal proxy Network Load Balancer
Cross-region internal proxy Network Load Balancer
Regional external passthrough Network Load Balancer 3
Regional internal passthrough Network Load Balancer 3

Load balancing aspects

To decide which load balancer best suits your implementation of Google Cloud, consider the following aspects of Cloud Load Balancing:

- Traffic type
- External versus internal load balancing
- Global versus regional load balancing
- Premium versus Standard Network Service Tiers
- Proxy versus passthrough load balancing

Traffic type

The type of traffic that you need your load balancer to handle is another factor in determining which load balancer to use.

Load balancer type	Traffic type
Application Load Balancers	HTTP or HTTPS
Passthrough Network Load Balancers	TCP or UDP These load balancers also support other IP protocol traffic such as ESP, GRE, ICMP, and ICMPv6.
Proxy Network Load Balancers	TCP with optional SSL offload

External vs Internal Load Balancing

- Google Cloud load balancers can be deployed as external or internal load balancers:
  - **External load balancers** distribute traffic that comes from the internet to your Google Cloud Virtual Private Cloud (VPC) network.
  - **Internal load balancers** distribute traffic that comes from clients in the same VPC network as the load balancer or clients connected to your VPC network by using VPC Network Peering, Cloud VPN, or Cloud Interconnect.

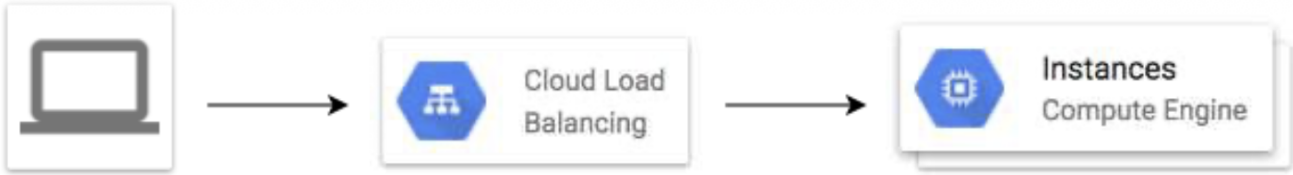
Global vs Regional Load Balancing

Load balancer scope	Architecture	Resilient to zonal outage	Resilient to regional outage
Global	Each load balancer is distributed across all regions	✓	✓
Cross-region	Each load balancer is distributed across multiple regions	✓	✓
Regional	Each load balancer is distributed across multiple zones in the region	✓	🚫 An outage in a given region affects the regional load balancers in that region

Proxy vs Passthrough Load Balancing

- Depending on your traffic type and client location (internal or external), you can choose between proxy or passthrough load balancers.
  - Proxy load balancers terminate client connections and create new ones to backends,
  - while passthrough load balancers forward packets unchanged to backend VMs, which handle connection termination and respond directly to clients.

Load Balancing - SSL/TLS Termination/Offloading



- Client to Load Balancer: Over internet
  - HTTPS recommended
- Load Balancer to VM instance: Through Google internal network
  - HTTP is ok. HTTPS is preferred.
- SSL/TLS Termination/Offloading
  - Client to Load Balancer: HTTPS/TLS
  - Load Balancer to VM instance: HTTP/TCP